

AI-Powered Analysis of Tourists' Opinions on Saudi Attractions

Artificial Intelligent · Sentiment Analysis · Tourist Opinions · Social Media Analytics

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01 INTRODUCTION

As Saudi Arabia's tourism sector continues to flourish, visitors' sentiments tell a compelling story. Our research explores tourist media coverage and online reviews to reveal how travellers experience the Kingdom's attractions, services, and hospitality. By examining these perceptions closely, it sheds light on both opportunities and shortcomings, offering tourism authorities valuable insight to elevate quality, enrich the visitor journey, and support the Kingdom's ambitious tourism future.

123M

visitors
Total 2025

%6.03

Growth
vs 2024

03 MODELS & EVALUATION MATRIX

ARABIC MODEL MARBERT

A BERT-based transformer pretrained on a large-scale Arabic corpus covering Modern Standard Arabic and diverse regional dialects. Fine-tuned on our tourism dataset to classify informal and dialect-rich tourist opinions.

Accuracy 87%
F1-score 92%

ENGLISH MODEL RoBERTa

A robustly optimized BERT-based transformer pre-trained on large-scale English corpora using dynamic masking and improved training strategies. Fine-tuned on our tourism dataset to classify English visitor reviews.

Accuracy 91%
F1-score 93%

Why two models?

Saudi tourism data is bilingual. Arabic requires a dialect-aware model (MARBERT) to handle informal expressions and code-switching, while English benefits from RoBERTa's robust contextual understanding. Together, they maximise sentiment classification accuracy across both languages.

TOOLS
USED



Python



power bi



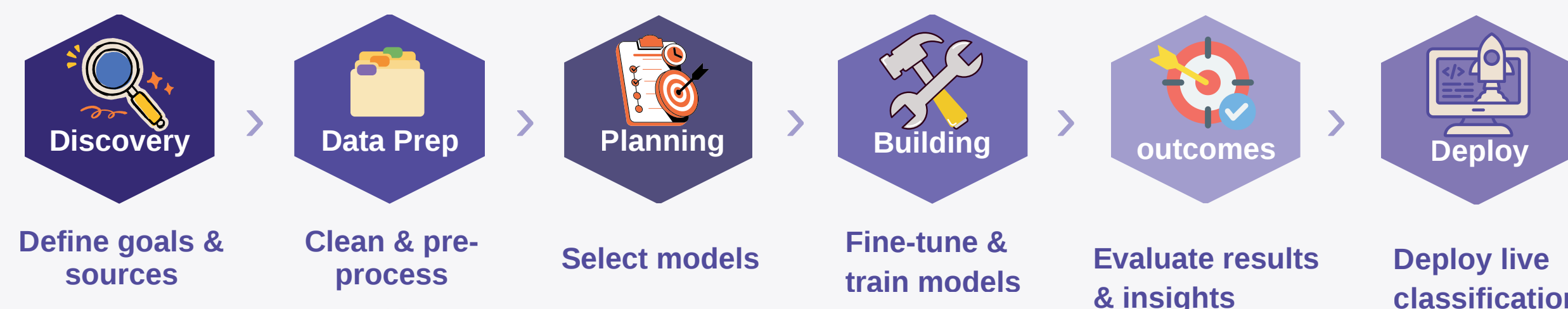
HTML



Excel

02 METHODOLOGY

DATA ANALYTICS LIFE CYCLE (DALC)



- 1 **Standardisation**
Lowercase, remove HTML, special chars & URLs
- 2 **Normalisation**
Remove Arabic diacritics, unify letter forms
- 3 **Stop-word Removal**
Discard high-frequency function words (AR + EN)
- 4 **Stemming / Lemmatisation**
Reduce words to root (e.g. رائعة -> رائع)
- 5 **Annotation**
Labels : ☒ positive ☐ Natural ☒ Negative

EXAMPLE TRANSFORMATION

ROW INPUT :
"زيارة العلاء كانت رائعة
جدا!#السياحة_السعودية"

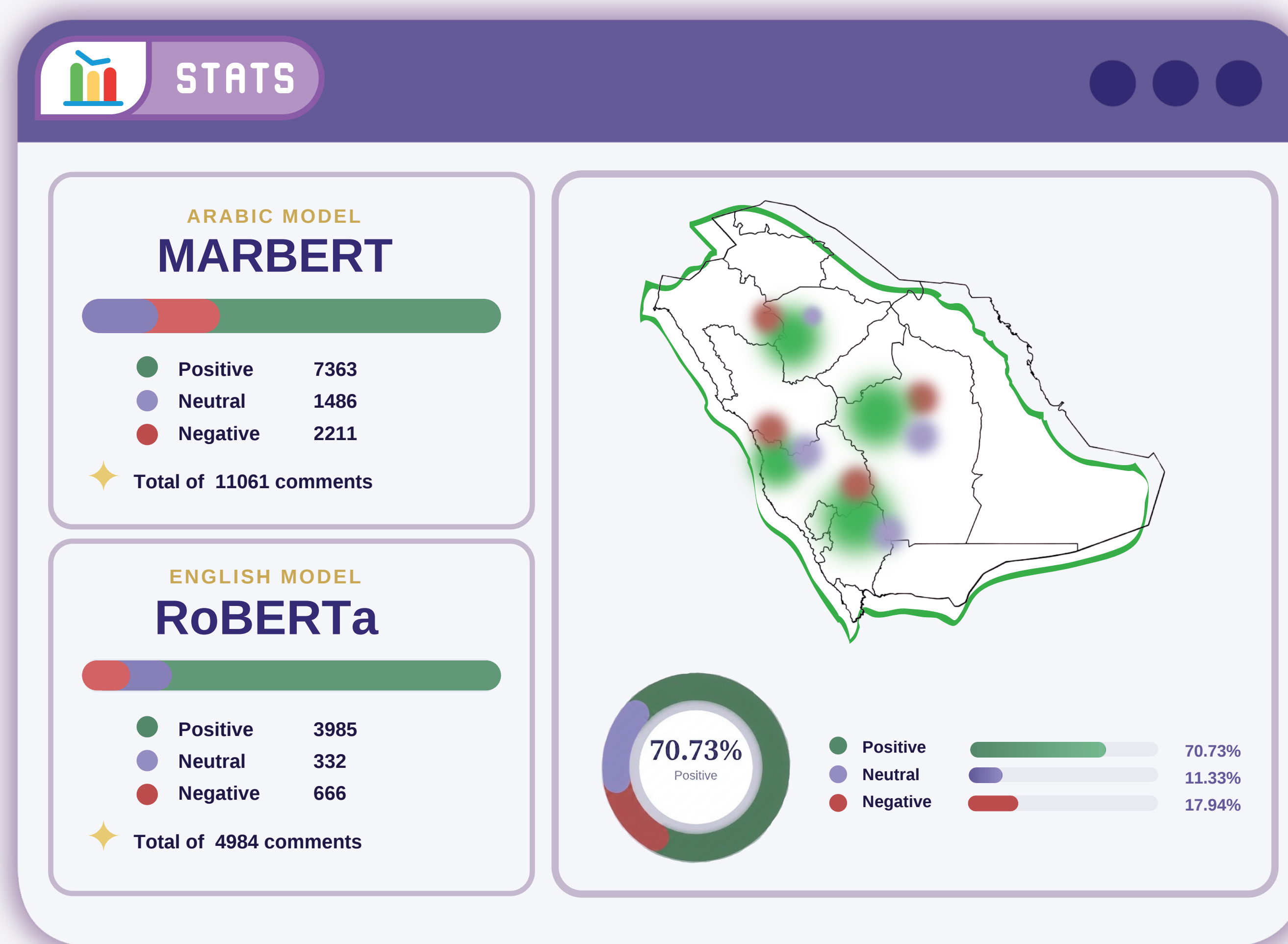
AFTER CLEAN :

زيارة العلاء رائع

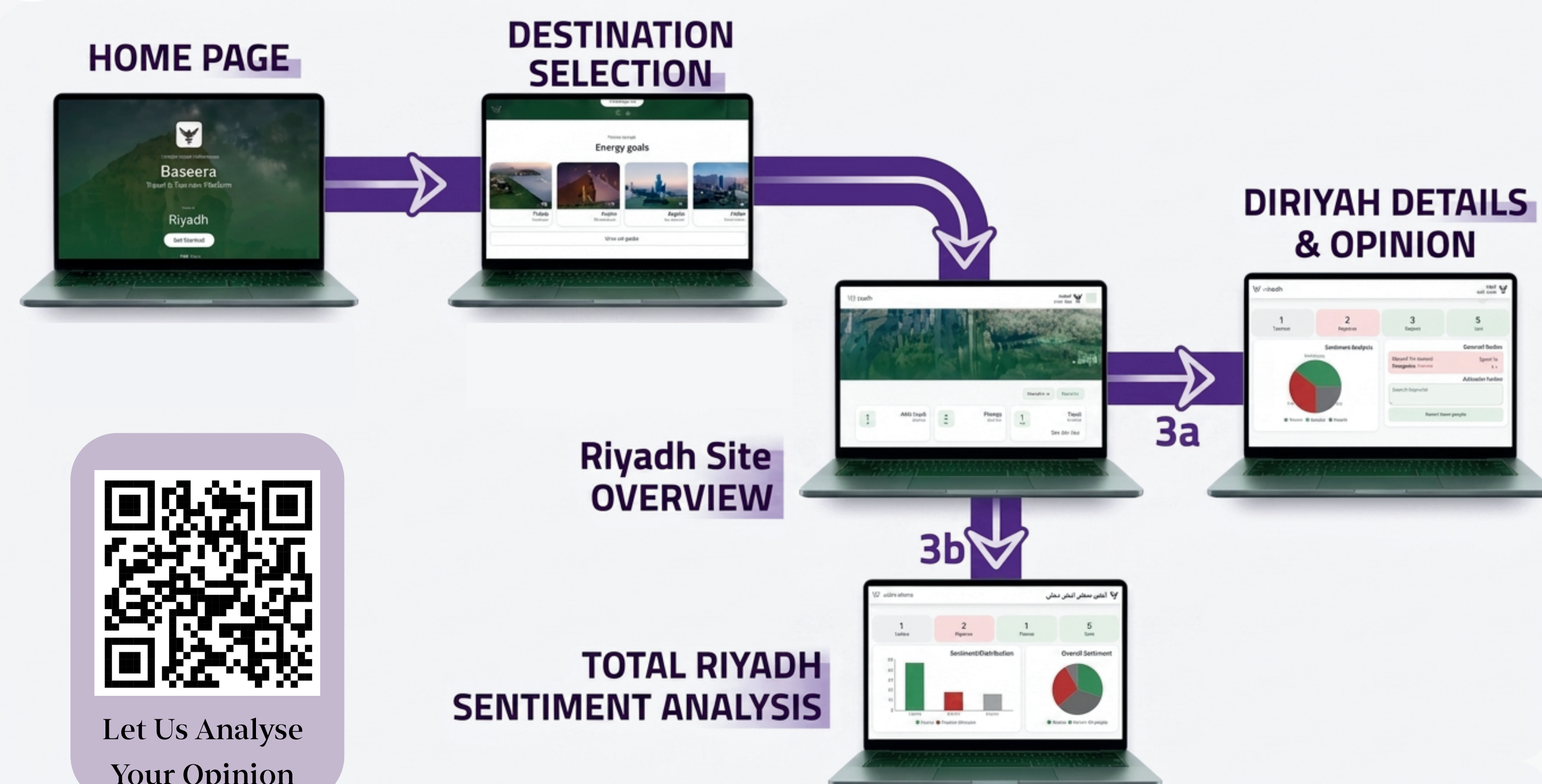
OUTPUT :

☒ Positive

04 RESULTS VISUALISATION DASHBOARD



05 USER INTERFACE DESIGN



06 CONCLUSION & FUTURE DIRECTIONS

This project showcases the effectiveness of AI in analysing tourist opinions, providing valuable insights for improving Saudi Arabia's tourism experience. The use of MARBERT and RoBERTa models has resulted in high classification accuracy, making it a reliable tool for sentiment analysis. Future work will focus on expanding the dataset to include more diverse data sources, such as video and real-time feedback. Further enhancement of multilingual capabilities will ensure broader accessibility. Additionally, real-time sentiment analysis and personalised recommendations could be explored. This approach contributes to Vision 2030 by supporting the growth of the tourism sector in Saudi Arabia.

